

FULL-LENGTH REPORT



Beyond screen time: The core influences of problematic screen use on adolescent development networks

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ABSTRACT

Background and aims: With surges in digital technologies, concerns over adolescents' screen use have intensified. Previous studies often relied on self-reported screen time, neglecting the experiential and motivational aspects of different screen activities (e.g. social media, gaming, and smartphones), possibly leading to heterogeneous associations. This study aimed to examine whether the severity of problematic screen use, conceptualized as a continuous measure of screen-related functional impairment, plays a more central role in development than self-reported screen time or phone-checking frequency, and to explore its influence within the broader adolescent ecosystem (i.e. family conflict, peer involvement, and school participation). **Methods:** Cohort data were obtained from the Adolescent Brain Cognitive Development (ABCD) study Release 5.1. Screen use was obtained from self-report questionnaires, capturing both activity time and functional impairments. Among cross-sectional networks derived from 9,054 youth (4,272 girls, 47.18%) at 2-year follow-up (T2, ages 11–12 years) and 4,007 youth (1,896 girls, 47.32%) at 4-year follow-up (T4, ages 13–14 years), problematic use showed higher centrality than screen time and checking behavior, owing to stronger connections with behavioral tendencies. Cross-lagged-panel-network analyses of problematic use included 3,954 youth (1,863 girls, 47.12%). **Results:** Problematic use exhibited high out-strength, which was associated with worsening psychopathologies and environmental conditions. Conversely, problematic use at T4 appeared less influenced by factors at T2. **Conclusion:** This study underscores the pivotal role of problematic screen use, which showed greater centrality and stronger predictive effects on adolescent well-being than self-reported screen time, highlighting the need for policies and interventions addressing screen-related functional impairments to promote healthier screen habits in developing youth.

KEYWORDS

problematic screen use, internet addiction, addictive behaviors, impulsive behaviors, adolescent development, network analysis

INTRODUCTION

Youth today are deeply immersed in digital environments, often dedicating substantial time to internet-based activities. In 2019, children aged 8 to 12 devoted more than 4 h daily to

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digital entertainment media, and teenagers spent nearly 7.5 h (Common Sense, 2022). The COVID-19-related school closures and remote learning activities further increased the use of digital technologies (Nagata, Cortez, et al., 2022). Given the prominence of screen use in the lives of children and adolescents, understanding its impact on development has become a critical research priority.

Despite the growing literature, findings on the impacts of screen use remain heterogeneous. Screen time, the duration of screen use, has been reported as exhibiting bidirectional relationships with negative developmental measures (Marciano, Ostroumova, Schulz, & Camerini, 2021), including externalizing (e.g. aggression, inattention) and internalizing (e.g. anxiety, depression) concerns (Eirich et al., 2022; Neville, McArthur, Eirich, Lakes, & Madigan, 2021; Riehm et al., 2019), behavioral problems (Song et al., 2023), and poor mental health and life satisfaction (Nagata, Cortez, et al., 2022; Orben, Przybylski, Blakemore, & Kievit, 2022; Santos et al., 2023). Poorer cognitive performance and differences in brain development have also been observed (Hutton, Dudley, Horowitz-Kraus, DeWitt, & Holland, 2020; Li et al., 2024; Yang et al., 2024). Nevertheless, other studies suggest potential adverse effects may be minimal (Santos et al., 2023) or explained by external factors, including family and school environments (Orben & Przybylski, 2019; Panayiotou, Black, Carmichael-Murphy, Qualter, & Humphrey, 2023). Notably, some positive measures, such as enhanced social connections during the pandemic (Marciano et al., 2021), have also been linked to screen use (Ophir, Rosenberg, Tikochinski, Dalyot, & Lipshits-Braziler, 2023; Sanders et al., 2024).

Meta-analyses suggest that the conflicting strength and direction of associations between screen use and mental health may, in part, stem from differences in how screen use is measured (Eirich et al., 2022; Purba et al., 2023; Tang, Werner-Seidler, Torok, Mackinnon, & Christensen, 2021). Many studies rely on subjective reporting of screen time and frequency (Kaye, Orben, Ellis, Hunter, & Houghton, 2020), which are susceptible to biases such as over- or under-estimation (Scharkow, 2016). While objective logs offer more precision (Hodes & Thomas, 2021; Scharkow, 2016), they fail to capture the experiential and motivational aspects of screen use (Odgers, Schueller, & Ito, 2020). These limitations highlight the importance of adopting multidimensional measures to capture the complex nature of screen use and its potential effects.

Beyond mere screen time, problematic screen use, characterized by persistent, compulsive patterns resistant to cessation and associated with functional impairment (Nagata, Singh, et al., 2022), may have more direct and clinically relevant associations with adverse outcomes. While problematic screen use is not currently recognized as a formal disorder in major diagnostic systems, its symptomatology shares similarities with other behavior addictions, such as Internet Gaming Disorder (IGD) in the Diagnostic and Statistical Manual of Mental Disorders Fifth Edition (DSM-5) and Gaming Disorder in the International Classification of Diseases 11th Revision (ICD-11), both of which emphasize impaired control, increased priority given to the behavior, and continuation despite negative consequences (American Psychiatric Association, 2013; World

Health Organization, 2018). Several self-report questionnaires are available to measure the problematic use severity, with or without a clinical threshold.

A study utilizing ecological momentary assessment found significant links between problematic usage of the internet and mental health concerns in individuals aged 12 to 23 years (Gansner, Nisenson, Carson, & Torous, 2020). Moreover, mediators such as sedentary behaviors, shorter sleep durations, and limited offline activities have been identified as pathways linking screen time and negative measures (Eirich et al., 2022; Khan, Lee, Rosenbaum, Khan, & Tremblay, 2021; Santos et al., 2023), further underscoring the significance of investigating problematic screen use.

Various types of screen use, such as social networking, gaming, and streaming videos, are additional factors to consider. The Interaction of Person-Affect-Cognition-Execution (I-PACE) model (Brand, Young, Laier, Wölfling, & Potenza, 2016, 2019), alongside empirical studies, suggests that adolescents' first-choice online activities may reflect behavioral tendencies and contribute differentially to developmental outcomes (Li et al., 2024; Wang et al., 2024; Yang et al., 2024). For example, individuals with high sensation-seeking often gravitate toward gaming (Hu, Zhen, Yu, Zhang, & Zhang, 2017). Such tendencies may be influenced by the unique content, interactive features, and goals associated with different screen-based activities. It is important to distinguish between types of screen activities to better understand their differential impacts on development.

The ecological systems theory emphasizes the important role of multiple environments in adolescents' mental and behavioral development, encompassing both risk and resilience factors (Bronfenbrenner & Ceci, 1994; Cross, 2017). Within this framework, microsystems, encompassing immediate environments directly influencing adolescents, may contribute proximally to development (Pantin, Schwartz, Sullivan, Coatsworth, & Szapocznik, 2003). This perspective has been applied to understanding the impact of screen use on youth development (Paulus et al., 2023). While studies indicate relatively minor impacts of overall screen time on adolescent development compared to influences like social relationships and life satisfaction (Orben & Przybylski, 2019; Panayiotou et al., 2023), certain patterns of screen use may exert more pronounced influences on adolescents.

Network analysis provides a powerful framework for exploring the organization and interaction of components within complex systems (Borsboom, 2017). Cross-sectional network analysis estimates unique relationships (edges) between components (nodes) while controlling for other variables, typically through partial correlations (Borsboom & Cramer, 2013). This approach identifies the centrality of nodes based on the strength of their connections, highlighting key constructs within the network. As illustrated in the lower left corner of Fig. 1, the Adolescent Brain Cognitive Development (ABCD) Study® offers a unique opportunity to investigate such relationships. By incorporating self-reported measures of screen time, problematic screen use, and other variables, the dataset allows for a comparative analysis of the centrality of screen use measures within networks.

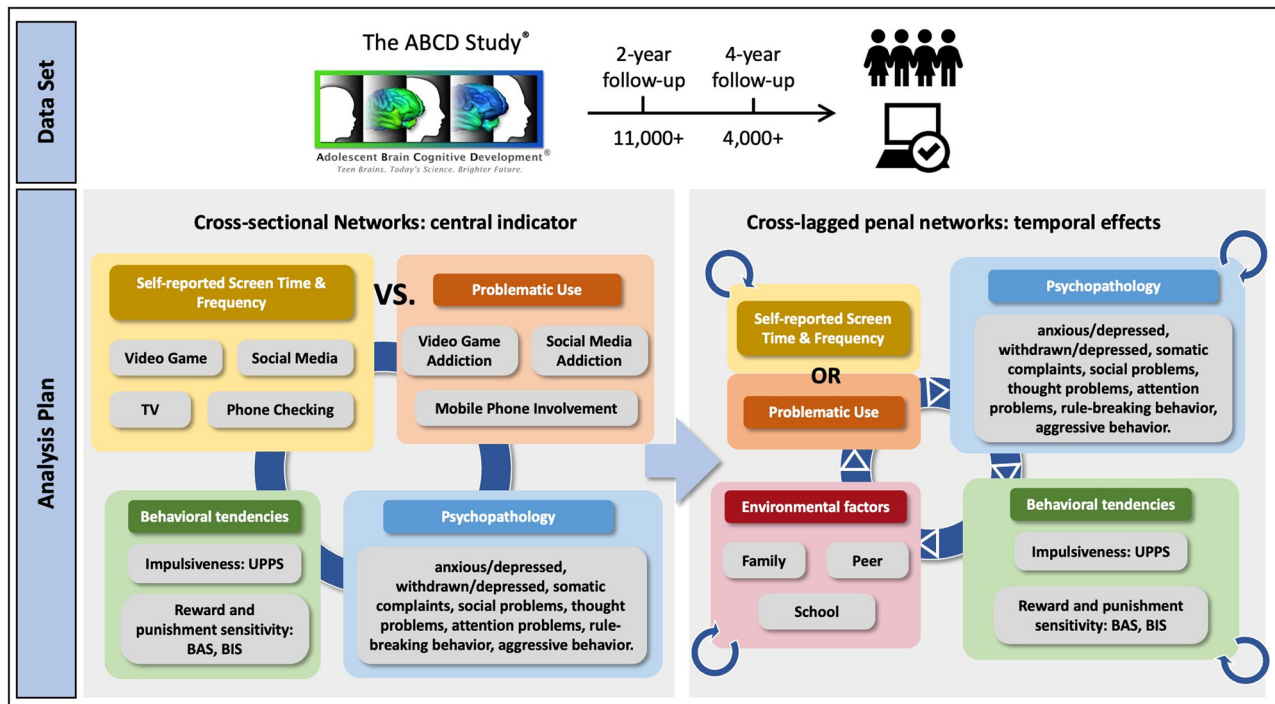


Fig. 1. Schematic diagram of the study design. The figure depicts the data set employed for analyses (top) and the analytic approach (bottom). The 2-year follow-up and 4-year follow-up of the ABCD study provided data (top). Data were incorporated into a cross-sectional network that included screen use measures, behavioral tendencies, and psychopathology, respectively, to identify central measures of screen use (bottom left). The core measures were then incorporated into longitudinal CLPN analyses along with behavioral tendencies, psychopathology, and environmental factors to examine and compare intertemporal effects (bottom right)

To further investigate the bidirectional relationships between screen use indicators identified as central and other developmental and psychopathological factors, cross-lagged-panel-network (CLPN) modeling was employed. CLPN modeling integrates cross-lagged panel models into networks, capturing interactions that occur within and between psychological constructs over time (Wysocki, Rhemtulla, Van Bork, & Cramer, 2022). Leveraging the ABCD Study dataset, which includes repeated measures (lower right corner of Fig. 1), this approach provides a robust framework for examining how screen use measures interact with broader developmental and environmental contexts across time (Borsboom et al., 2021; Epskamp, 2020).

METHOD

Participants

The participants were drawn from ABCD Study® Release 5.1. Detailed information regarding the study's design and recruitment strategies can be found elsewhere (Garavan et al., 2018). Participants who completed both 2-year follow-up (T2, ages 11–12 years, 2018–2020) and 4-year follow-up (T4, ages 13–14 years, 2020–2022) were included to construct separate cross-sectional networks, and those who participated in T4 were included in the CLPN. To ensure the independence of the sample, one individual was randomly selected from families with multiple youth participants. Demographic

characteristics for the subsample are presented in Table 1, and an analytic flowchart is provided in eFig. 1.

Variables

The study variables related to screen usage, behavioral and psychopathological factors, and environmental factors are outlined below, with additional information in supplementary materials.

Screen usage. Screen usage measures include self-report screen time and problematic usage. The Screen Time Questionnaire (STQ) assesses the duration of youth engagement in recreational activities. Based on responses to drop-down lists, the average daily screen time devoted to (1) video games, (2) social media (including social networking, texting, and video chatting), and (3) TV shows or movies were aggregated. The mobile phone checking frequency was obtained from the Mobile Phone Attachment assessment. Additionally, problematic video game, social media, and mobile phone use were assessed using the Video Game Addiction Questionnaire (VGAQ), the Social Media Addiction Questionnaire (SMAQ), and the Mobile Phone Involvement Questionnaire (MPIQ) (Andreassen, Torsheim, Brunborg, & Pallesen, 2012; Walsh, White, & Young, 2010). Although the scale measures used in this study are not directly based on the mainstream diagnostic criteria for behavioral addiction, they capture overlapping dimensions (see Supplementary eTables 3 and 4). We treat problematic screen use as continuous variables to capture the full spectrum of screen usage (or severity of problematic

Table 1. Demographics of the study sample

	2-year-follow-up <i>n</i> = 9,054			4-year-follow-up <i>n</i> = 4,007			2-year-follow-up to 4-year-follow-up <i>n</i> = 3,954		
	Female <i>n</i> = 4,272	Male <i>n</i> = 4,782	<i>p</i> -value	Female <i>n</i> = 1,896	Male <i>n</i> = 2,111	<i>p</i> -value	Female <i>n</i> = 1,863	Male <i>n</i> = 2,091	<i>p</i> -value
Age at T2, mean (SD), m	144.12 (8.01)	144.28 (7.96)	0.343	/	/	/	143.21 (7.77)	143.11 (7.72)	0.681
Age at T4, mean (SD), m	/	/	/	168.66 (7.89)	168.78 (8.05)	0.612	168.53 (7.88)	168.63 (8.13)	0.681
Race, No. (%)			0.153			0.101			0.038
White	2,712 (63%)	3,127 (65%)		1,281 (68%)	1,498 (71%)		1,250 (67%)	1,491 (71%)	
Black	663 (16%)	709 (15%)		222 (12%)	224 (11%)		225 (12%)	216 (10%)	
Asian	119 (2.8%)	107 (2.2%)		51 (2.7%)	59 (2.8%)		52 (2.8%)	55 (2.6%)	
Mixed/other	778 (18%)	839 (18%)		342 (18%)	330 (16%)		336 (18%)	329 (16%)	
Ethnicity, No. (%)			0.718			0.506			0.565
Non-Hispanic	3,378 (79%)	3,796 (79%)		1,493 (79%)	1,644 (78%)		1,469 (79%)	1,633 (78%)	
Hispanic	894 (21%)	986 (21%)		403 (21%)	467 (22%)		394 (21%)	458 (22%)	
Parents marital status, No. (%)			0.434			0.201			0.152
Married or living with a partner	3,136 (73%)	3,545 (74%)		1,425 (75%)	1,623 (77%)		1,400 (75%)	1,612 (77%)	
Single	1,136 (27%)	1,237 (26%)		471 (25%)	488 (23%)		463 (25%)	479 (23%)	
Parental highest education, No. (%)			0.942			0.704			0.784
High school or less	1,137 (27%)	1,276 (27%)		451 (24%)	513 (24%)		443 (24%)	505 (24%)	
College education	3,135 (73%)	3,506 (73%)		1,445 (76%)	1,598 (76%)		1,420 (76%)	1,586 (76%)	
Family income, No. (%)			0.731			0.845			0.965
<\$25,000	737 (17%)	833 (17%)		281 (15%)	323 (15%)		274 (15%)	319 (15%)	
\$25,000–\$49,999	616 (14%)	652 (14%)		280 (15%)	285 (14%)		266 (14%)	291 (14%)	
\$50,000–\$74,999	517 (12%)	618 (13%)		248 (13%)	274 (13%)		245 (13%)	269 (13%)	
\$75,000–\$99,999	637 (15%)	692 (14%)		288 (15%)	314 (15%)		291 (16%)	311 (15%)	
\$100,000–\$199,999	1,308 (31%)	1,486 (31%)		584 (31%)	680 (32%)		580 (31%)	670 (32%)	
\$200,000+	457 (11%)	501 (10%)		215 (11%)	235 (11%)		207 (11%)	231 (11%)	

Note. Two sample *t*-test for continuous variables; Pearson's chi-squared test for categorical variables.

usage) and its varying degrees of impact on adolescent development. We also conducted sensitivity analyses using categorical indicators of problematic use, which are presented in the [supplementary materials](#).

Psychopathology. Caregivers provided annual reports on youth's psychopathological symptoms starting from baseline using the Child Behavior Checklist (CBCL) (Achenbach & Rescorla, 2006). Domains included anxious/depressed, withdrawn/depressed, somatic complaints, social problems, thought problems, attention problems, rule-breaking behavior, and aggressive behavior.

Behavioral tendencies. Behavioral tendencies are defined as consistent patterns of thoughts, feelings, or behaviors that individuals exhibit in response to specific situations or stimuli. These tendencies can evolve, shaped by both environmental and personal factors. In this study, impulsivity and behavioral inhibition/activation system were measured bi-annually by the brief urgency, perseverance, premeditation, and sensation seeking (UPPS) scale (Watts, Greene, Bonifay, & Fried, 2023) and BIS/BAS scale (Carver & White, 1994; Pagliaccio et al., 2016).

Environmental factors. Family, school, and peer measures are included as environmental factors. The Conflict subscale in the Family Environment Scale (FES) gauged conflict levels within family settings. Two subscales from the Peer Behavior Profile (PBP), Prosocial Peer Involvement (PPI) and Delinquent Peer Involvement (DPI) provided insights into peer interactions. Three subscales from the School Risk and Protective Factors (SRPF), namely perceived school environment, positive school involvement, and school disengagement, were included.

Data preprocessing

Truncation. Following the weighted average, very few outliers were identified relative to daily screen time, and these were truncated to the maximum limit of 23 h and 45 min set by drop-down list.

Imputation. Addressing missing data in screen use measures necessitated careful consideration to avoid biased inclusion. Some missing values in problematic usage scales were due to participants not meeting prerequisites for specific activities, such as not playing video games, lacking social media accounts, or not owning mobile devices. These instances were imputed with the minimum scale value, representing a self-reported “never.” The remaining missing data, satisfying random missingness as per Little's MCAR test (eFigs 2–4), were imputed using Full Information Maximum Likelihood in cross-sectional networks and predictive mean matching in CLPN. To ensure the robustness of the results, we also conducted analyses using pairwise deletion for missing values arising from the absence of certain screen activities. The sensitivity analysis results are provided in the [supplementary materials](#), demonstrating the stability of findings across different imputation approaches.

Reversed scoring. Regarding PPI, perceived school environment, and positive school involvement were considered protective factors for individual development. To ensure consistency with other risk factors in the network analysis, scoring was reversed.

Covariates. The analysis incorporated age, race, ethnicity, parental marital status, parental highest education, and family income as covariates due to their significant associations with screen usage (Trinh et al., 2020) and individual psychopathological factors. Standardized residuals, accounting for covariate effects, were utilized in subsequent analyses.

Network estimation

Cross-sectional network. A Gaussian graphical model (GGM) was constructed using the Extended Bayesian Information Criterion (EBIC) in combination with graphical lasso (gLASSO) algorithms (Epskamp, Borsboom, & Fried, 2018). This approach leveraged penalty parameters to achieve sparsity and identify the optimal set of neighboring factors for each node. Edge thickness corresponded to the partial correlation coefficient, denoting the association between nodes while controlling for all other variables.

CLPN analyses. Central indicators of screen use were integrated into CLPN models along with psychopathological and environmental factors. Autoregressive coefficients in CLPN models reflect the predictive power of a node at T2 on itself at T4, adjusted for all other nodes at T2. Cross-lagged coefficients indicate the predictive power of a node at T2 on a different node at T4, adjusted for all other nodes at T2. A LASSO-penalized maximum-likelihood procedure, with 10-fold cross-validation for tuning parameter selection, was applied to shrink small coefficients to zero.

Centrality estimation

In cross-sectional networks, centrality was assessed using strength and expected influence (EI). Strength sums the absolute weights of edges connecting a node to all other nodes, while EI sums the weights of edges, considering their direction (positive or negative). Given the directed nature of edges in CLPNs, in-strength, out-strength, in-EI, and out-EI were calculated to evaluate the influence of a node on others (out) and the influence received from others (in). For the Screen Usage nodes, centrality was determined by considering both intra-community connections within the Screen Usage cluster and inter-community connections with other nodes. Since the aim of this study was to explore the relationships between screen usage measures and other developmental factors, potential community effects were addressed by incorporating bridge strength as an auxiliary centrality index. This helps capture the strength of connections across different communities. The nonparametric bootstrap method was utilized to assess the robustness and differences in node centrality and edge strength. The R packages *qgraph* (Epskamp, Cramer, Waldorp, Schmittmann, & Borsboom, 2012), *bootnet* (Epskamp & Fried, 2018) *glmnet* (Friedman,

Hastie, & Tibshirani, 2010), and *NetworkComparisonTest* (Van Borkulo, 2017) facilitated these analyses. Given sex differences in screen media activity (Su, Han, Yu, Wu, & Potenza, 2020) in a manner that relates differently to mental health during adolescence (Paulus et al., 2023) and changes in types of screen media consumed from childhood to adolescence that differ by sex (Song et al., 2023), we considered sex-specific models.

Ethics

The coordinating center, located at the University of California, obtained ethical approval of the ABCD protocols. All caregivers provided written informed consent and all children provided assent to participate in the study.

RESULTS

Sex-specific cross-sectional networks

As shown in Fig. 2A, across all four networks comprising 18 nodes, female networks and T4 networks exhibited sparser connectivity compared to male networks and T2 networks

(T2: female, 98/153, 64.05%; male, 103/153, 67.32%; T4: female, 83/153, 54.25%; male, 97/153, 63.40%). Network comparisons showed no significant difference between networks at the same time point and between T2 and T4 networks of the same sex (all $ps > 0.12$, eFigs 5 and 6). Given the well-documented sex differences in adolescent screen use and relationships with health measures in existing research (Paulus et al., 2023), subsequent analyses were conducted separately by sex. The correlation stability (CS) coefficients showed the stability of networks (all CSs > 0.5 , eFig. 7).

More central nodes in cross-sectional networks

The nonparametric bootstrapped difference test showed problematic use exhibited greater strength and EI compared to corresponding screen time and checking behavior (Fig. 2C and D). Across all networks, problematic mobile phone use (A3) demonstrated significantly larger strengths (T2: Female, 1.20 vs. 0.91, 95% CI = [0.148, 0.349], Male, 1.20 vs. 0.89, 95% CI = [0.174, 0.366]; T4: Female, 1.10 vs. 0.87, 95% CI = [0.124, 0.415], Male, 1.20 vs. 0.79, 95% CI = [0.265, 0.537]) and EI (T2: Female, 1.10 vs. 0.74, 95% CI = [0.262, 0.483], Male, 1.10 vs. 0.74, 95% CI = [0.292,

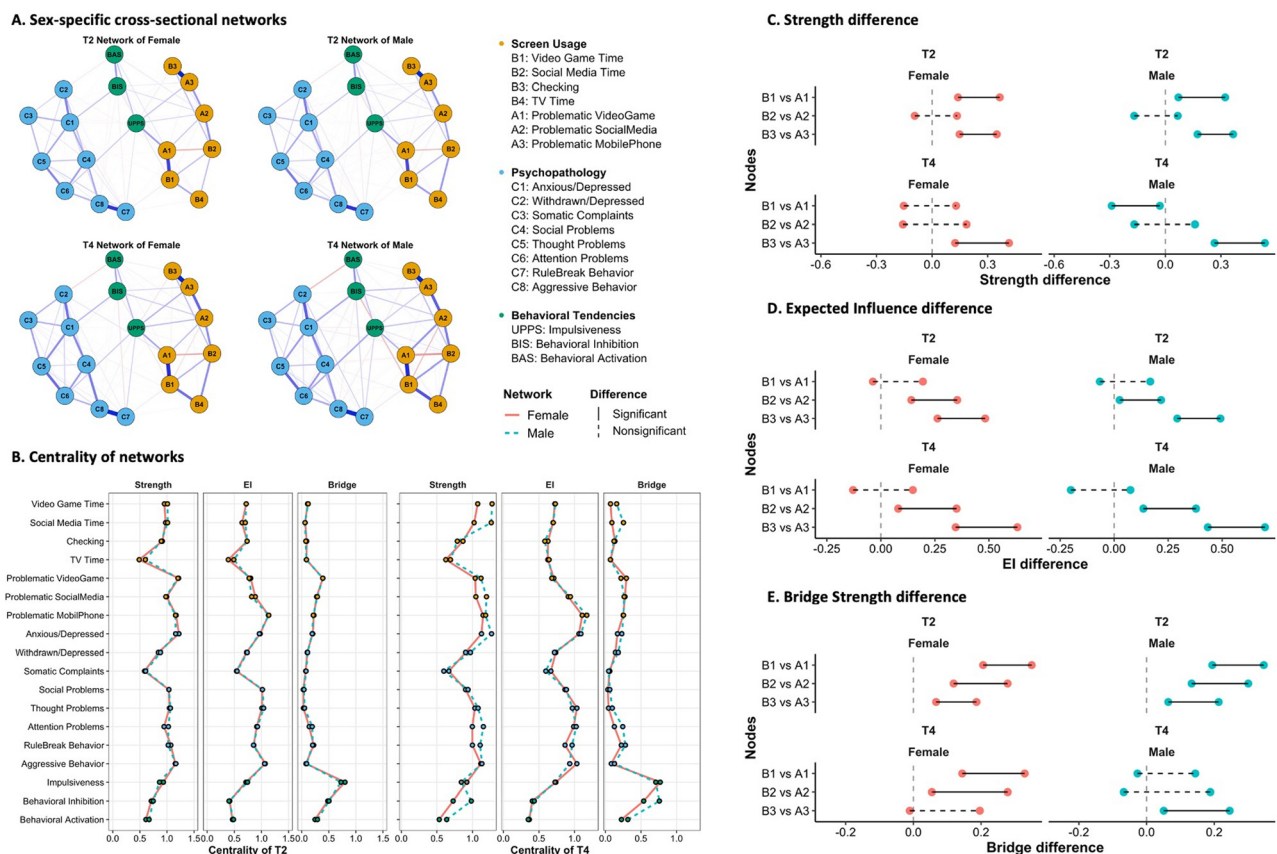


Fig. 2. Sex-specific cross-sectional networks and their centrality indices. (A) Edge thickness indicates the strength of the association, and color indicates the positive (green) and negative (red) association. Node color indicates the community of nodes. For enhanced clarity in visualizing sex-specific networks concurrently, an average layout technique was employed. (B) Centrality indexes of female (red solid line) and male (blue dashed line) network nodes are presented separately. The point color indicates the community of nodes. (C, D, E) The results of the nonparametric bootstrapped difference tests are presented. Points with different colors (red = female, blue = male) represent 95% CI values of node centrality indicators in the same networks. Solid lines (significant) and dashed lines (nonsignificant) show interval ranges

0.492]; T4: Female, 1.10 vs. 0.63, 95% CI = [0.346, 0.632], Male, 1.20 vs. 0.58, 95% CI = [0.434, 0.698]) compared to checking behavior (B3). Problematic video game use (A1) displayed higher strengths in T2 networks (T2: Female, 1.20 vs. 0.95, 95% CI = [0.140, 0.365], Male, 1.20 vs. 1.00, 95% CI = [0.071, 0.323]) and lower strengths in T4 male networks (1.10 vs. 1.30, 95% CI = [−0.290, −0.029]) compared to video game time (B1). However, there was no significant difference in EI (T2: Female, 0.80 vs. 0.72, 95% CI = [−0.037, 0.195], Male, 0.77 vs. 0.72, 95% CI = [−0.066, 0.167]; T4: Female, 0.72 vs. 0.71, 95% CI = [−0.129, 0.148], Male, 0.68 vs. 0.73, 95% CI = [−0.200, 0.075]) between video game time and problematic use. Similarly, problematic social media use (A2) exhibited greater EI (T2: Female, 0.89 vs. 0.64, 95% CI = [0.141, 0.353], Male, 0.82 vs. 0.70, 95% CI = [0.025, 0.217]; T4: Female, 0.91 vs. 0.70, 95% CI = [0.080, 0.350], Male, 0.94 vs. 0.70, 95% CI = [0.135, 0.379]) in all networks compared to social media time (B2), with no significant difference in strengths (T2: Female, 0.99 vs. 0.97, 95% CI = [−0.093, 0.134], Male, 0.82 vs. 0.70, 95% CI = [−0.168, 0.067]; T4: Female, 1.00 vs. 1.00, 95% CI = [−0.157, 0.186], Male, 1.20 vs. 1.30, 95% CI = [−0.168, 0.160]). The centrality of some screen use nodes was significantly different between T2 and T4, but it did not affect the centrality of problematic use nodes compared with screen time nodes (eTable 9). The results demonstrate the stability of findings across different imputation approaches (eFig. 17). Some screen use duration or checking behavior frequency indicators still exhibited significantly higher centrality compared to categorical counterparts (eFig. 20).

Strong associations were observed between screen usage measures (Fig. 2A and eTable 9), prompting further comparison of nodes' bridging strength (Fig. 2D). Problematic usage exhibited higher bridge strength in T2 networks for both females (video game: 0.39 vs. 0.11, 95% CI = [0.206, 0.350]; social media: 0.28 vs. 0.06, 95% CI = [0.119, 0.278]; mobile phone: 0.23 vs. 0.09, 95% CI = [0.067, 0.186]) and males (video game: 0.38 vs. 0.12, 95% CI = [0.194, 0.346]; social media: 0.29 vs. 0.06, 95% CI = [0.133, 0.300]; mobile phone: 0.21 vs. 0.08, 95% CI = [0.064, 0.213]) compared to corresponding screen time and checking behavior. At T4, the bridge strength of problematic video game (0.29 vs. 0.07, 95% CI = [0.144, 0.329]) and social media (0.28 vs. 0.09, 95% CI = [0.054, 0.279]) use among females was higher than corresponding screen time, while the bridge strength of problematic mobile phone use among males (0.25 vs. 0.12, 95% CI = [0.051, 0.246]) significantly exceeded checking behavior.

Furthermore, significant differences were observed across content for the same indicator types. Screen time for watching TV consistently exhibited the lowest values across centrality indexes. Pairwise comparative significance of all node centrality is described in supplementary materials (eFigs 9–11).

Closer associations with behavioral tendencies lead to higher centrality

Although problematic usage indicators generally exhibited greater centrality compared to their counterparts, it did not

necessarily imply stronger edges to all nodes outside the community. Within each network, the strongest connections were typically observed within the community (eTable 8). Additionally, connections between problematic use and behavioral tendencies were often stronger.

Nonparametric bootstrapped difference tests on edge strength between paired nodes (Fig. 3) revealed that problematic social media and mobile phone use showed significantly stronger positive associations with impulsivity and behavioral inhibition across all networks. In contrast, video game time had a stronger negative association with behavioral inhibition. Problematic video game and social media use also demonstrated notably positive associations with behavioral activation compared to corresponding nodes in T2 networks. Conversely, in edges involving psychopathological nodes, no significant differences were found between nodes with different screen use measures, except for a stronger positive connection between social media time and rule-breaking behaviors.

Sex-specific CLPNs

In light of these findings, problematic usages, as central measures of screen use, were incorporated into CLPN analyses. All factors exhibited self-influence from the previous time point, with problematic social media use and problematic mobile phone use displaying weaker autoregressive effects compared to problematic video game use (Fig. 4A). Autoregressive effects (mean B: female = 0.406, male = 0.391) overshadow cross-lagged effects (mean B: female = 0.012, male = 0.013), shaping the depiction of cross-lagged paths in visualization. Consequently, autoregressive paths were set to 0 to accentuate cross-lagged effects most pertinent to our research objectives (Fig. 4B). Structural similarity between the two networks was evaluated using Pearson's product-moment correlation, revealing a significant positive correlation ($r = 0.47$, $t = 10.382$, $p < 0.001$, 95% CI [0.39, 0.55]). The CS coefficients underscored acceptable network stability (eFig. 12).

The impact of problematic screen use on individual development

Within numerous cross-lagged edges (female: 198/380, 52.11%; male: 205/380, 53.95%), our focus primarily centers on the edges involving problematic screen use. Across both networks, problematic screen nodes exhibited notable centrality (Fig. 4C). All nodes of problematic screen use demonstrated higher out-strength, signifying their substantial influence on other nodes, albeit with lower in-strength. After isolating effects within the community, bridge strength underscored the centrality of problematic use. Problematic use significantly impacted psychopathology, behavioral tendencies, and environmental factors. Conversely, other factors exerted less influence on problematic use (eTable 10). Sensitivity analysis verified the stability of these results (eFigs 18 and 21).

Among females, heightened problematic social media and mobile phone use were significantly associated with adverse outcomes at T4. Higher problematic social media use at T2 correlated with increased somatic complaints (A2→C3,

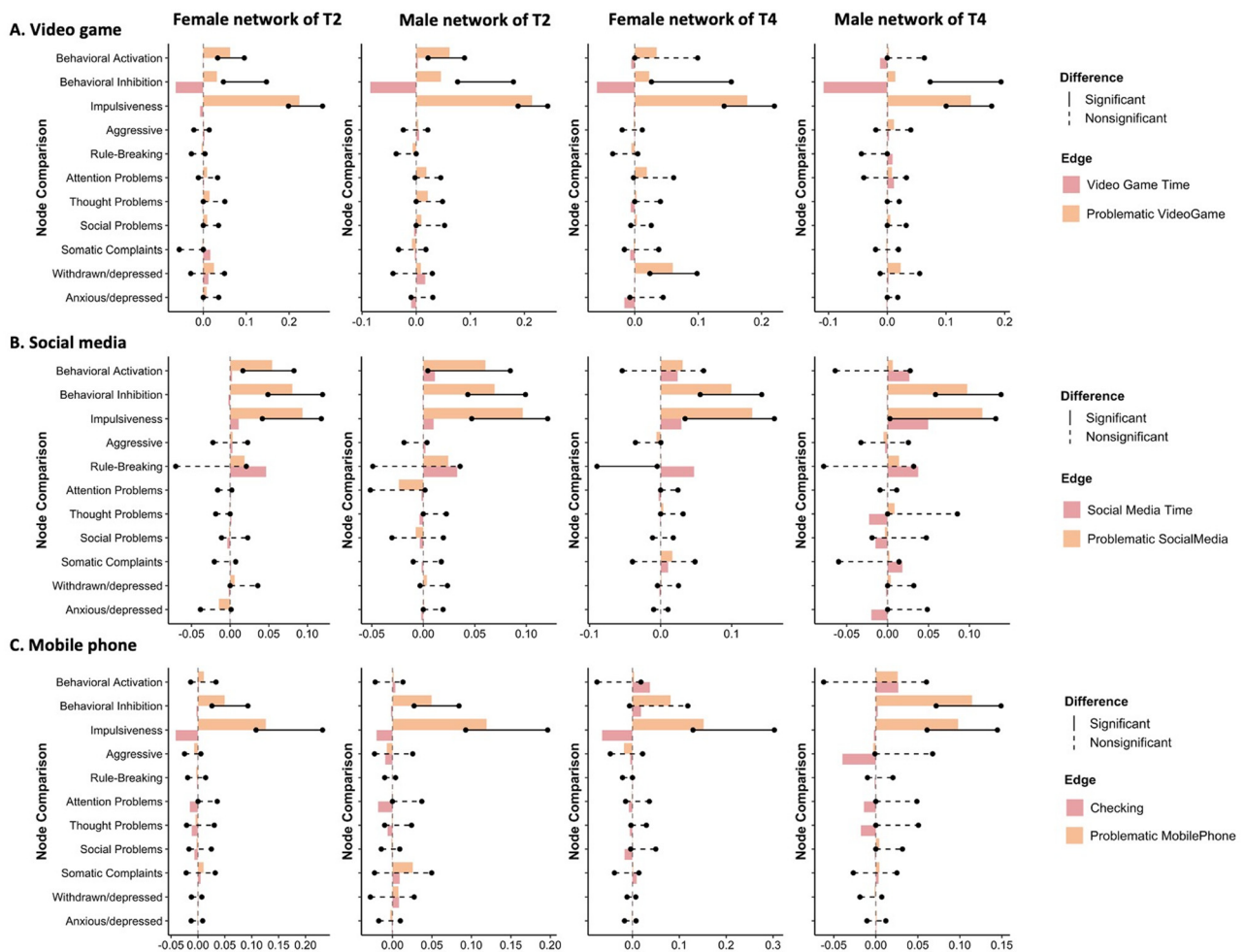


Fig. 3. The nonparametric bootstrapped difference test of edges. The bar chart shows the edge strength between screen usage nodes and other nodes outside the community. Points represent 95% CI values of node centrality indicators in the same networks. Solid (significant) and dashed lines (nonsignificant) show interval ranges

$B = 0.302$), behavioral activation ($A2 \rightarrow BAS$, $B = 0.179$), delinquent peer involvement ($A2 \rightarrow E3$, $B = 0.108$), and fewer thought problems ($A2 \rightarrow C5$, $B = -0.251$) at T4. Problematic mobile phone use was associated with heightened impulsivity ($A3 \rightarrow UPPS$, $B = 0.219$), behavioral activation ($A3 \rightarrow BAS$, $B = 0.194$), behavioral inhibition ($A3 \rightarrow BIS$, $B = 0.108$) and diminished school involvement ($A3 \rightarrow E5$, $B = 0.108$), and adverse school environments ($A3 \rightarrow E4$, $B = 0.100$). Additionally, a bidirectional association was observed between problematic mobile phone and social media use ($A2 \rightarrow A3$, $B = 0.118$; $A3 \rightarrow A2$, $B = 0.110$). Conversely, heightened problematic video game use correlated with fewer somatic complaints ($A1 \rightarrow C3$, $B = -0.357$), fewer withdrawn/depressed symptoms ($A1 \rightarrow C2$, $B = -0.248$), fewer attention problems ($A1 \rightarrow C6$, $B = -0.229$), less behavioral activation ($A1 \rightarrow BAS$, $B = -0.214$), and less behavioral inhibition ($A1 \rightarrow BIS$, $B = -0.186$).

In the male network, problematic use similarly exerted effects on other domains. Specifically, heightened problematic social media use correlated with more behavioral activation ($A2 \rightarrow BAS$, $B = 0.315$), impulsivity ($A2 \rightarrow UPPS$, $B = 0.308$), rule-breaking problems ($A2 \rightarrow C7$, $B = 0.178$),

behavioral inhibition ($A2 \rightarrow BIS$, $B = 0.133$), social problems ($A2 \rightarrow C4$, $B = 0.120$), anxiety/depression symptoms ($A2 \rightarrow C1$, $\beta = 0.113$), and attention problems ($A2 \rightarrow C6$, $B = 0.139$). Elevated problematic mobile phone use was also associated with more impulsivity ($A3 \rightarrow UPPS$, $B = 0.149$) and behavioral activation ($A3 \rightarrow BAS$, $B = 0.115$). A bidirectional relationship was observed between problematic mobile phone and social media use ($A2 \rightarrow A3$, $B = 0.137$; $A3 \rightarrow A2$, $B = 0.108$). Conversely, heightened problematic video game use correlated with lower impulsivity ($A1 \rightarrow UPPS$, $B = -0.197$), less behavioral inhibition ($A1 \rightarrow BIS$, $B = -0.153$), fewer withdrawn/depressed symptoms ($A1 \rightarrow C2$, $B = -0.152$), fewer attention problems ($A1 \rightarrow C6$, $B = -0.121$), and less prosocial peer involvement ($A1 \rightarrow E2$, $B = 0.121$).

DISCUSSION

It is critical to understand the impact of screen usage on adolescent development and move beyond screen time. This

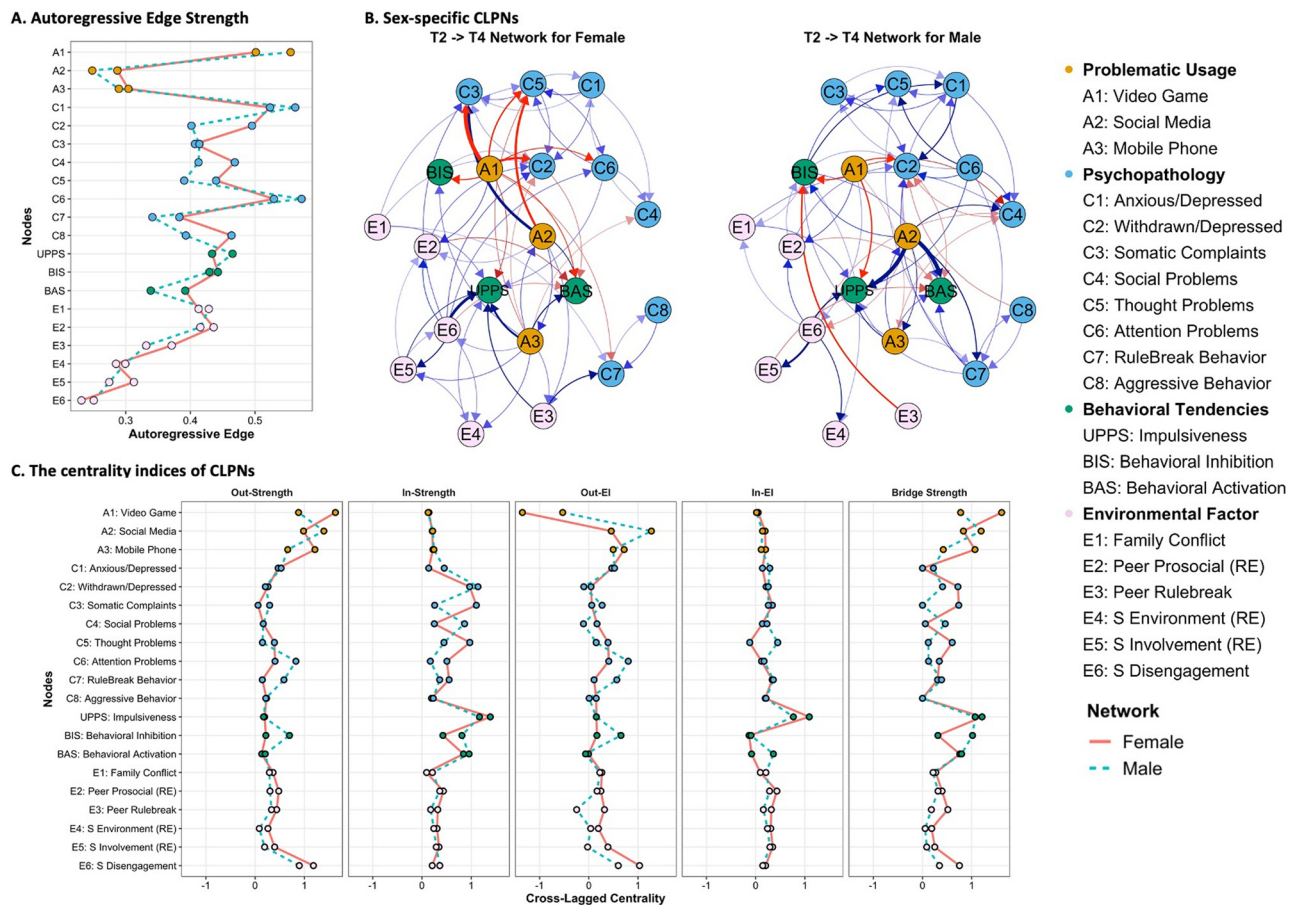


Fig. 4. Sex-specific CLPNs and their autoregressive edge strength. (A) The autoregressive coefficients for each node in the female (red solid line) and male (blue dashed line) networks are presented separately. (B) In the networks, the arrow of the edge represents the direction of the association, the thickness represents the strength of the association, and the color represents a positive (blue) or negative (red) association. Node placement was determined based on the average strength of sex-specific networks. Autoregressive edges, weaker edges (i.e. B within ± 0.05), and covariates were excluded from the plot to ease visual interpretation. (C) Centrality indices of female (red solid line) and male (blue) network nodes are presented separately

study, utilizing data from the ABCD study, shows that problematic screen use indicators exhibited greater centrality in developmental networks compared to screen time and checking behaviors and were strongly associated with psychopathology and behavioral tendencies, indicating their potential as key factors in adolescent development. To further explore their role in adolescent development, CLPN analysis was conducted, encompassing psychopathology, behavioral tendencies, and environmental factors. The analyses revealed that problematic screen use at T2 influenced other factors at T4, but not vice versa, indicating problematic use has a significant impact on adolescent development over time.

Problematic screen use: a key factor

Problematic screen use emerged as a central element within adolescent developmental networks. Cross-lagged analyses further revealed that problematic use at T2 predicted environmental factors and psychopathological symptoms at T4. These findings underscore the critical importance of

focusing on indicators of problematic screen use, which go beyond the mere measurement of screen time. Problematic screen use incorporates functional impairments offering a clinically relevant perspective (Ellis, 2019). While screen time has been linked to cognitive and emotional functioning, examining problematic use allows for a deeper understanding of the maladaptive mechanisms that underlie these associations. The effects of usage patterns (e.g. parental guidance), content (e.g. educational content for children; Wang et al., 2024), and usage scenarios (e.g. television-viewing during family meals; Yang et al., 2024) suggest that considering only screen time is insufficient for capturing full impacts on adolescent development.

The distinction between screen time and problematic screen use is crucial for understanding their respective developmental effects. While an association between problematic use and screen time seems reasonable, as evidenced by cross-sectional networks in this study, prolonged screen time does not necessarily indicate functional impairments (Ellis, Davidson, Shaw, & Geyer, 2019). Functional impairment reflects poorly controlled usage patterns of screens that

intrude upon and disrupt daily life, potentially leading to tolerance and withdrawal (Ioannidis, Grant, & Chamberlain, 2022). Furthermore, there is a bidirectional relationship between screen time and the degree of problematic use, where increased time spent on specific screen activities may reflect a deeper and potentially harmful engagement with these activities. The relationship between psychopathological symptoms and screen time may not always translate into problematic use. For example, while rule-breaking behaviors were correlated with gaming time in cross-sectional networks, this relationship did not extend to problematic gaming. This highlights the need to differentiate between mere usage and the functional impairments that characterize problematic use in order not to overly pathologize common recreational activities while at the same time preventing harmful effects on youth development.

Interplay of behavioral tendencies and problematic screen use

Behavioral tendencies, particularly impulsivity and behavioral inhibition/activation, appear to play a crucial role in the relationships between problematic screen use and psychopathological symptoms. For instance, empirical studies have highlighted the predictive role of social media use in attention deficits, with impulsivity identified as a potential risk factor for symptoms of attention-deficit hyperactivity disorder (Thorell, Burén, Ström Wiman, Sandberg, & Nutley, 2024). This suggests that impulsivity may act as a mediator in the relationship between excessive social media use and psychopathological outcomes. Adolescence has been proposed as a period of developmental imbalance, where reward and emotional circuits mature faster than prefrontal control mechanisms (Giedd, 2008). This imbalance increases susceptibility to impulsivity and reward-seeking behaviors, which may contribute to addiction-like patterns of screen use (González-Bueso et al., 2018; Jordan & Andersen, 2017; Shulman et al., 2016). This study supports these theories, revealing robust associations and significant cross-lagged effects between impulsivity, reward-seeking, and problematic screen use. While the influence of other factors on problematic screen use was relatively weak, these findings emphasize the need for further investigation into the factors that shape screen usage and its association with developmental outcomes.

Impact varies by screen content

The varying impacts of screen content were also evident in this study. Although television-viewing constituted considerable a significant portion of screen time, it showed weaker associations with behavioral and psychopathological outcomes compared to more interactive screen use (e.g. gaming or social media). Motivational differences, such as escapism in gaming or social validation in social media, may explain these discrepancies. For example, motives for using social media were differentially linked to mental health outcomes longitudinally during the pandemic, with coping motives linked to higher COVID-19-related distress and problematic usage and social motives linked to improvements from

distress (Buodo, Moretta, Santucci, Chen, & Potenza, 2023; Moretta, Buodo, Santucci, Chen, & Potenza, 2023). Gaming also may alter reward-punishment sensitivity and contribute to maladaptive behaviors over time. Conversely, regulated content on television, such as educational programs, may mitigate adverse effects, underscoring the role of content moderation in shaping developmental outcomes (Committee on Public Education, 2001).

Adolescence is also marked by heightened social demands and sensitivity, rendering social media and mobile phone use central in adolescent networks (Crone & Konijn, 2018; Madigan & Reich, 2023). Contrary to previous assumptions that online social interaction primarily affects females, our study reveals problematic social media and mobile phone use significantly relate to greater behavioral activation, externalizing and internalizing symptoms, social challenges, and attention concerns in both sexes. Despite boys typically exhibiting lower levels of social media and phone use than girls, the impacts may be substantial for both boys and girls. Additionally, problematic video game use may blunt adolescents' general reward-punishment sensitivity, which may be due to the rewarding characteristics of video games and the dissimulation of reward in addictive disorders (Yao, Zhang, Fang, Liu, & Potenza, 2022).

Navigating the digital landscape: beyond time limits

Even when considering peer, family, and school environments, problematic screen use remains a pivotal factor, suggesting that regulating digital technology use may not be a Sisyphean Cycle (Orben, 2020). Recent guidelines, such as limiting recreational screen time to no more than two hours per day, have been recommended to safeguard adolescents' physical health and encourage social engagement. In China, for example, regulations restrict video game usage for juveniles to just one hour on non-school days (Buckley, 2021). In Australia, recent legislative efforts have striven to restrict social media use to individuals under the age of 16 years (Bobby Allyn, 2024). However, based on the present findings, functional impairments resulting from screen use appear to be a more direct influence on adolescent development than screen time alone. While time limits may be a practical "second-best" policy approach, focusing on strategies that prevent functional damage due to excessive screen use—while also fostering responsible technology use—should be prioritized for promoting healthy development in adolescents.

Moreover, compared to the substantial difference in centrality between problematic use and screen time observed in T2, this difference diminished over time by T4. Although the impact of screen time may be relatively minor, it is important to recognize that small streams may contribute to the formation of seas. Thus, the cumulative effects of screen use should be considered, particularly given that screen time tends to increase with age.

Limitations and strengths

Limitations warrant consideration. First, the ABCD study has constraints. Some factors relevant to screen use and

adolescents, such as emotional regulation tendencies, were not included due to limitations in available variables. Additionally, only the T2 and T4 data met the study's criteria, with T4 data limited to a subset of participants and potentially influenced by the COVID-19 pandemic, complicating interpretations. While the longitudinal sample aligns with the cross-sectional sample on key variables, larger samples and longer follow-ups would enhance robustness.

Regarding problematic screen use, while modeling it as a continuum provides greater statistical power, the use of clinically relevant thresholds should also be considered. Our analyses showed consistent patterns using both continuous and thresholded measures, reinforcing the validity of our findings. Although the measurement scales in this study are not directly derived from major diagnostic frameworks, such as the DSM-5 and ICD-11, they encompass overlapping dimensions, capturing key aspects of problematic screen use relevant to behavioral addiction criteria. Future research should integrate standardized diagnostic criteria to enhance the clinical relevance of findings, particularly when distinguishing between different types of screen activities, such as gaming and social media use. Given the relatively limited scope of screen use among early adolescents, further investigation is warranted to elucidate factors influencing development and persistence of problematic use over extended timelines.

Finally, for both the cross-sectional and longitudinal networks, the observed centrality and associations reflect correlations rather than causal effects. Further research is needed to explore potential causal relationships between screen use indicators and developmental measures.

Despite limitations, this study provides valuable insights into the differential impacts of various types and patterns of screen use. By employing both cross-sectional and cross-lagged network analysis, this study provides a more nuanced understanding of screen use impacts on adolescents, with data supporting problematic usage exerting subsequent impacts on the mental health of developing youth. While existing research emphasizes the importance of screen time, this study highlights the need to consider factors such as usage motivation, content type, and interactions with real-world environments. The findings have important implications for policy-making and interventions aimed at promoting healthy screen use among adolescents.

CONCLUSION

In summary, the findings of this study underscore the centrality of problematic screen use in adolescent development, surpassing the relevance of screen time or phone-checking frequency. Problematic screen use, characterized by functional impairments, demonstrated stronger associations with behavioral tendencies and adverse developmental outcomes. These associations suggest problematic screen use exerts significant influence on adolescent psychopathologies and

environmental conditions. The study also revealed that while problematic screen use at a later stage (T4) is less influenced by earlier factors (T2), its impact on well-being can be profound. Consequently, interventions and policies should prioritize addressing functional impairments related to screen use to foster healthier screen habits and mitigate adverse developmental effects among adolescents.

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Conflict of interest: The authors declare no conflict of interest with the content of this manuscript. MNP discloses that he has consulted for and advised Game Day Data, Addiction Policy Forum, Boehringer Ingelheim, BariaTek, and Opiant Therapeutics; been involved in a patent application involving Novartis and Yale; received research support from the Mohegan Sun Casino, Children and Screens and the Connecticut Council on Problem Gambling; consulted for or advised legal, non-profit and gambling entities on issues related to impulse control, internet use and addictive behaviors; performed grant reviews; edited journals/journal sections; given academic lectures in grand rounds, CME events and other clinical/scientific venues; and generated books or chapters for publishers of mental health texts; serves as an associate editor of the Journal of Behavioral Addictions.

Data and code availability: All analytic codes for results in this article are available at https://github.com/LynnXu96/Network_ABCDstudy.

Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive DevelopmentSM (ABCD) Study (<https://abcdstudy.org>), held in the NIMH Data Archive (NDA). This is a multisite, longitudinal study designed to recruit more than 10,000 children age 9–10 and follow them over 10 years into early adulthood. The ABCD Study[®] is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089,

U24DA041123, U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at https://abcdstudy.org/consortium_members/. ABCD consortium investigators designed and implemented the study and/or provided data but did not participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or ABCD consortium investigators. The ABCD data repository grows and changes over time. The ABCD data used in this report came from NDA DOI: 10.15154/z563-zd24. DOIs can be found at <https://nda.nih.gov/study.html?id=2313>.

SUPPLEMENTARY MATERIAL

Supplementary data to this article can be found online at <https://doi.org/10.1556/2006.2025.00035>.

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